Entity Relationship Model (ERM) Proposal

ChargeAlaya - EV Charging Payment and Management System

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Executive Summary

ChargeAlaya is a complete Electric Vehicle Charging Payment and Management System that aims at making the electric vehicle charging process more straightforward to electric vehicle owners, charging station operators and system administrators. This proposal is an Entity Relationship Model that defines the entire database design, entities, attributes, relations, and constraints required to support the functionality of the system.

The system that is proposed handles user accounts, vehicle details, charging station details, charger details and specifications, reservation booking, charging session, payment, customer feedback and maintenance tracking. The database is developed according to the industry best practices so that there is a guarantee of data integrity, scalability, and optimum performance. The design is normalized to Third Normal Form, as well as being useful in a real-world application.

Keywords: ChargeAlaya, Electric Vehicle Charging Payment and Management System, Entity Relationship Model, database design, user accounts, charging station details, charger details, reservation booking, payment, data integrity, scalability, Third Normal Form

1. Project Overview

1.1 Purpose

There are three user groups that ChargeAlaya supports. EV Owners are able to search stations, book chargers, control vehicles, pay. Station Owners control the stations, reservations, charging session, and maintenance. Administrators can monitor the work of systems, control users and actions on the platforms.

1.2 Scope

The system is compatible with secure authentication of users and role-based access control, vehicle registration and control, and full inventory of charging stations and chargers with real-time availability. It has a reservation system with conflict prevention, real-time monitoring of charging sessions with energy and cost monitoring, multi-method payment processing of eSewa, Khalti, card, cash, user feedback with the ratings and reviews, maintenance reporting and tracking.

1.3 Technology Stack

The system utilizes the following technologies:

- **Database Management System:** MySQL 5.7+
- **Storage Engine:** InnoDB (ACID compliance, foreign key support)
- **Character Set:** UTF8MB4 (Unicode support)
- **Backend:** PHP 7.4+
- **Frontend:** Bootstrap 5.3, JavaScript ES6

2. Software Architecture

ChargeAlaya adheres to 3-Tier Architecture, which supports separation of concerns, maintainability and independent scalability.

Clear image available at: ChargeAlaya Software Architecture.png

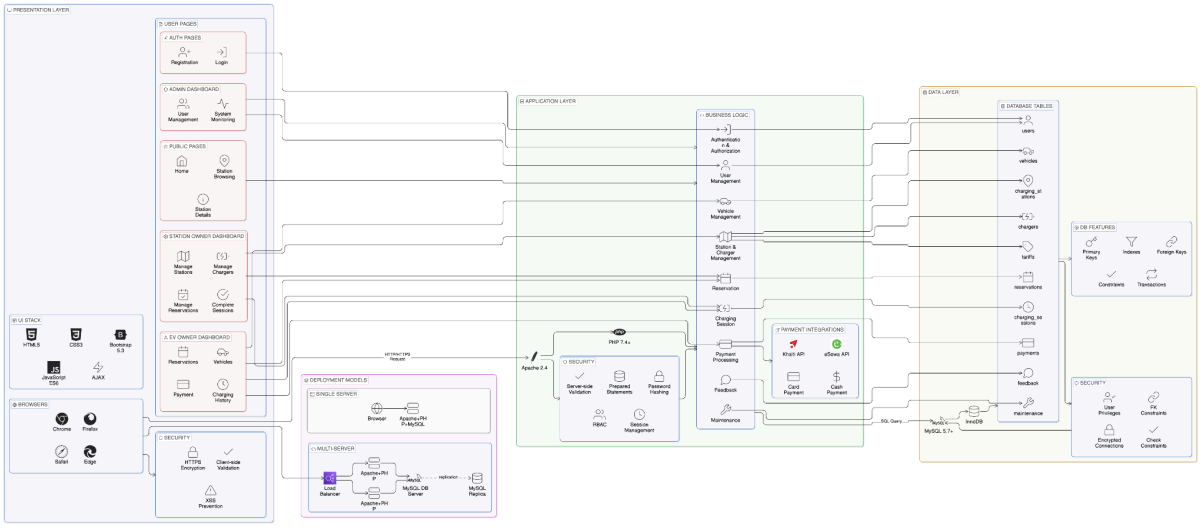

Fig. 1: ChargeAlaya Software Architecture

2.1 Presentation Tier

Presentation Tier offers a responsive web interface based on HTML5, CSS3, Bootstrap 5.3 and JavaScript ES6. It manages user used interaction, form input, client side validation and response rendering.

2.2 Application Tier

Application Tier It is an Application that is developed as PHP 7.4+ over Apache 2.4 and includes core business logic. It handles authentication, user and vehicle management, station and charger operations, reservation, charging sessions and payments, feedback and maintenance. Such layer implements the business-rules in time consistency, connector consistency, conflict resolution and role accessibility.

2.3 Data Tier

MySQL 5.7+ with InnoDB is used in the Data Tier to hold 10 normalized tables that include users, vehicles, stations, chargers, tariffs, reservations, sessions, payments, feedback and maintenance. Consistency and ACID transactions are guaranteed by referential integrity, indexing, and ACID transactions.

2.4 Inter-Tier Communication in Three-Tier Architecture

The tier communication is request-response based whereby the HTTP/HTTPS requests move between the Presentation Tier and Application Tier and SQL activities are managed by the Data Tier.

2.5 Security Architecture (Defense-in-Depth Approach)

There is security applied on every level: HTTPS and client-side validation at the Presentation Tier, server-side validation, bcrypt password hashing, prepared statements, role enforcement at the Application Tier and privilege control with constraints at the Data Tier. The Application Tier integrates with external payment gateways (eSewa and Khalti) that are secure.

2.6 External System Integration and Modularity

External intersections with payment gateway (eSewa and Khalti) are integrated with Applications Tier to enable the system to handle online payments safely without any connection with business logic. This modular design would provide that payment processing can be revised or expanded without having an impact on other parts of the system.

3. Business Requirements

3.1 Functional Requirements

3.1.1 User Management

User Management module deals with user registering, assigning roles (Admin, Station Owner, EV Owner), and secure authentications with the help of bcrypt. Users are able to maintain and update their profiles and the administrators are able to close down or block accounts.

3.1.2 Vehicle Management

Vehicle Management enables the EV owners to add more vehicles with details like battery capacity, connector type and unique license plate identification.

3.1.3 Charging Station Management

The Charging Station Management allows the station owners to control the various stations, change or update the location, working hours, availability and introduce the station specific tariff.

3.1.4 Charger Management

Charger Management keeps separate records of the status of individual chargers per station such as charger type (Fast, Normal), power rating, connector type and real-time availability.

3.1.5 Tariff Management

The Tariff Management allows the station owners to establish as well as to vary the prices they impose on their service to the ev owners.

3.1.6 Reservation Management

Reservation System offers booking in terms of time with a start and end time, tracking of the lifecycle, and avoidance of conflicting reservations.

3.1.7 Charging Session Management

The Session Management of charging captures the actual charging events, and it monitors the energy usage and calculates the costs properly.

3.1.8 Payment Management

Payment Processing entails the eSewa, Khalti, card and cash payments, transaction time stamp and tracking (pending, paid).

3.1.9 Feedback Management

Feedback System gathers 1-5 star feedback and customer feedback on the quality of the service.

3.1.10 Maintenance Management

Maintenance Management will provide the possibility to report issues, track their status (open, in progress, resolved) and monitor their resolution in order to be sure of the chargers reliability.

3.2 Non-Functional Requirements

Fulfillment of the non-functional requirements are mentioned below,

- Foreign keys, check constraints and default values are used to ensure **Data Integrity**.
- Strategic indexing of the common query attributes is used to optimize **performance**.
- **Scalability** is also in the form of normalized design and modular architecture.
- To avoid SQL injection, **security** is implemented through hashing passwords with bcrypt hash and role-based access control and prepared statements.

4. Entity Relationship Model

4.1 Entities

ChargeAlaya system has ten central entities that are represented by different roles in the system architecture.

- **Users** save information of every type (administrators, station owners, EV owners) and allow authentication, role management and access control.
- **Vehicles** contains EV information possessed by users, including an ability to match charging history to connector type and store vehicle-specific history.
- **Charging Stations** are physical points of charging, which facilitate location-based searches and management of stations by its owners.
- **Chargers** stores descriptions and real-time position of distinct charging stations. A variety of chargers with different levels of power and connector types are supported in a single station.
- **Tariffs** sets the prices of the stations, such as per-kWh, and fixed service fees, which enables flexible and competitive pricing models.
- **Reservations** is used to deal with booking charging slots, time allocation, preclude conflicts, and track the reservation lifecycle.

- **Charging Sessions** are used to track a real charge activity where the energy used is recorded, and the usage is based on actual use to charge the customer.
- **Payments** hold the financial transaction information associated with sessions such as payment method, time when it was placed, and payment status.
- **Feedback** records user ratings and comments on stations to help monitor the quality of the service and help users make an informed choice.
- **Maintenance** tracks the problems with chargers and logs reports and status development (open, in progress, resolved) to attain the equipment reliability.

4.2 Attributes

4.2.1 Users Entity

Attribute	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each user
first_name	VARCHAR(100)	NOT NULL	User's first name
last_name	VARCHAR(100)	NOT NULL	User's last name
email	VARCHAR(255)	UNIQUE, NOT NULL	User's email address (used for login)
password_hash	VARCHAR(255)	NOT NULL	Hashed password using bcrypt
phone	VARCHAR(10)	NOT NULL	Contact phone number
role	ENUM('admin', 'station_owner', 'ev_owner')	NOT NULL, DEFAULT 'ev_owner'	User's role in the system
status	ENUM('active', 'blocked')	DEFAULT 'active'	Account status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp

Table 1: Data Dictionary - Users Entity

4.2.2 Vehicles Entity

Attribute	Data Type	Constraints	Description
vehicle_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each vehicle
user_id	INT	FOREIGN KEY --> users(user_id), NOT NULL	Owner of the vehicle
brand	VARCHAR(100)	NOT NULL	Vehicle manufacturer
model	VARCHAR(100)	NOT NULL	Vehicle model name
battery_capacity_kwh	DECIMAL(6,2)	NULL	Battery capacity in kilowatt-hours
connector_type	VARCHAR(50)	NULL	Compatible connector (CCS2, CHAdeMO, Type2)
license_plate	VARCHAR(50)	NULL	Vehicle registration number
manufacturing_year	INT	NULL	Year of manufacture

Table 2: Data Dictionary - Vehicles Entity

4.2.3 Charging Stations Entity

Attribute	Data Type	Constraints	Description
station_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each station
station_name	VARCHAR(200)	NOT NULL	Name of the charging station
user_id	INT	FOREIGN KEY --> users(user_id), NOT NULL	Station owner
address	TEXT	NOT NULL	Full physical address
city	VARCHAR(100)	NULL	City location
province	VARCHAR(100)	NULL	Province/state
latitude	DECIMAL(10,8)	NULL	GPS latitude coordinate
longitude	DECIMAL(11,8)	NULL	GPS longitude coordinate
operating_hours	VARCHAR(100)	NULL	Station operating hours
status	ENUM('online', 'offline', 'under_maintenance')	DEFAULT 'online'	Current station status

Table 3: Data Dictionary - Charging Stations Entity

4.2.4 Chargers Entity

Attribute	Data Type	Constraints	Description
charger_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each charger
station_id	INT	FOREIGN KEY --> charging_stations(station_id), NOT NULL	Parent station
charger_type	ENUM('fast', 'normal')	NOT NULL	Type of charger
max_power_kw	DECIMAL(6,2)	NOT NULL	Maximum power output in kilowatts
connector_type	VARCHAR(50)	NULL	Connector type (CCS2, CHAdeMO, Type2)
status	ENUM('available', 'in_use', 'maintenance')	DEFAULT 'available'	Current charger status

Table 4: Data Dictionary - Chargers Entity

4.2.5 Tariffs Entity

Attribute	Data Type	Constraints	Description
tariff_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each tariff
station_id	INT	FOREIGN KEY --> charging_stations(station_id), NOT NULL	Associated station
service_fee	DECIMAL(10,2)	NULL	Fixed service charge per session
price_per_kwh	DECIMAL(10,2)	NOT NULL	Cost per kilowatt-hour
peak_start_time	TIME	NULL	Peak pricing start time
peak_end_time	TIME	NULL	Peak pricing end time

Table 5: Data Dictionary - Tariffs Entity

4.2.6 Reservations Entity

Attribute	Data Type	Constraints	Description
reservation_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each reservation
user_id	INT	FOREIGN KEY --> users(user_id), NOT NULL	Customer making reservation
charger_id	INT	FOREIGN KEY --> chargers(charger_id), NOT NULL	Reserved charger
vehicle_id	INT	FOREIGN KEY --> vehicles(vehicle_id), NULL	Vehicle for charging
start_time	DATETIME	NOT NULL	Reservation start time
end_time	DATETIME	NOT NULL	Reservation end time
status	ENUM('confirmed', 'cancelled', 'completed')	DEFAULT 'confirmed'	Reservation status

Table 6: Data Dictionary - Reservations Entity

4.2.7 Charging Sessions Entity

Attribute	Data Type	Constraints	Description
session_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each session
user_id	INT	FOREIGN KEY --> users(user_id), NOT NULL	Customer
vehicle_id	INT	FOREIGN KEY --> vehicles(vehicle_id), NOT NULL	Vehicle being charged
charger_id	INT	FOREIGN KEY --> chargers(charger_id), NOT NULL	Charger used
start_time	DATETIME	NOT NULL	Session start time
end_time	DATETIME	NULL	Session end time
energy_used_kwh	DECIMAL(10,2)	NULL	Actual energy consumed
cost	DECIMAL(10,2)	NULL	Final calculated cost
session_status	CHAR(10)	NOT NULL, DEFAULT 'completed'	Session status

Table 7: Data Dictionary - Charging Sessions Entity

4.2.8 Payments Entity

Attribute	Data Type	Constraints	Description
payment_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each payment
session_id	INT	FOREIGN KEY --> charging_sessions(session_id), NOT NULL	Associated charging session
user_id	INT	FOREIGN KEY --> users(user_id), NOT NULL	User making payment
amount	DECIMAL(10,2)	NOT NULL	Payment amount
payment_method	ENUM('esewa', 'khalti', 'card', 'cash', 'pending')	DEFAULT 'pending'	Selected payment method
transaction_time	DATETIME	DEFAULT CURRENT_TIMESTAMP	Payment timestamp
payment_status	ENUM('pending', 'paid')	DEFAULT 'pending'	Payment status

Table 8: Data Dictionary - Payments Entity

4.2.9 Feedback Entity

Attribute	Data Type	Constraints	Description
feedback_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each feedback
user_id	INT	FOREIGN KEY --> users(user_id), NOT NULL	User providing feedback
station_id	INT	FOREIGN KEY --> charging_stations(station_id), NOT NULL	Station being reviewed
rating	INT	CHECK (rating >= 1 AND rating <= 5)	Star rating (1-5)
comment	TEXT	NULL	Feedback text
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Feedback submission time

Table 9: Data Dictionary - Feedback Entity

4.2.10 Maintenance Entity

Attribute	Data Type	Constraints	Description
maintenance_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each maintenance record
charger_id	INT	FOREIGN KEY --> chargers(charger_id), NOT NULL	Charger with issue
reported_by	INT	FOREIGN KEY --> users(user_id), NOT NULL	User who reported the issue
issue_description	TEXT	NULL	Description of the problem
reported_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	When issue was reported
fixed_date	DATETIME	NULL	When issue was resolved
status	ENUM('open', 'in_progress', 'resolved')	DEFAULT 'open'	Current maintenance status

Table 10: Data Dictionary - Maintenance Entity

4.3 Relationships

4.3.1 One-to-Many Relationships

Multiple Vehicles are owned by users (1:N). There are many vehicles registered by an EV owner, which, however, are owned by a single user. Cascade delete deletes user vehicles.

There are several Charging Stations which are managed by users (1:N). The owner of a station can own several stations but one station has a single owner. Cascade delete gets rid of stations belonging to the owner.

Chargers are found in Charging Stations (1:N). A station contains several points of charge each charge point is part of just one station. Cascade delete is used to delete chargers with station.

Multiple Reservations are booked by users (1:N). Users use multiple reservation, one reservation by a user. The cascade delete deletes the reservations of the user.

There are several Reservations on vehicles (1:N). A vehicle contains several reservations, each reservation is optional and is made to one vehicle. SET NULL does not destroy history in the case of vehicle deletion.

There are a variety of Reservations (1:N) in Chargers. A charger can be reserved by several time slots with each slot having one reservation in the charger. Cascade delete can delete reservations having charger.

Multiple Charging Sessions are generated by the users (1:N). Users share a number of sessions, and each session is of one user. Cascade delete eliminates the sessions of the user.

There are several Charging Sessions of vehicles (1:N). There are various sessions of a vehicle and each session uses a vehicle. Cascade delete gets rid of vehicle sessions.

There are multiple Sessions (1:N) with the use of a Charger. The usage of a charger is in various sessions, with one session per charger. Cascade delete is used to delete sessions containing charger.

Payments are made several times by the users (1:N). The users make several payments, and a single payment is made by one user. Cascade delete is one that deletes payments of the user.

There are several Feedback (1:N) by users. Feedbacks are written by the users in a multi-feedback form. Cascade delete eliminates the feedback of user.

Several Feedback (1:N) are in stations. Stations have several feedbacks, one feedback per station. When feedback with station is deleted, the station is removed too.

There are several Maintenance (1:N) needed by Chargers. There are various maintenance records of a charger, records of each charger. Cascade delete gets rid of maintenance with charger.

There are several Maintenance (1:N) reported by users. Users complain of several problems, and each of them is reported by one of the users. Cascade delete deletes user reports.

4.3.2 One-to-One Relationships

There is a single Tariff (1:1) in Charging Stations. Every tariff is valid on a single station, every tariff will be based on one of the stations. Cascade delete deletes station tariff.

Charging Sessions consist of a single payment (1:1). One record of payment is produced by each session, and each payment is one session. Cascade delete deletes session Based payment.

4.4 Cardinality Summary

Relationship	Entity 1	Cardinality	Entity 2
OWNS	Users	1:N	Vehicles
MANAGES	Users	1:N	Charging Stations
CONTAINS	Charging Stations	1:N	Chargers
HAS	Charging Stations	1:1	Tariffs
BOOKS	Users	1:N	Reservations
RESERVES FOR	Vehicles	1:N	Reservations
RESERVED BY	Chargers	1:N	Reservations
INITIATES	Users	1:N	Charging Sessions
CHARGED AT	Vehicles	1:N	Charging Sessions
USED IN	Chargers	1:N	Charging Sessions
GENERATES	Charging Sessions	1:1	Payments
MAKES	Users	1:N	Payments
PROVIDES	Users	1:N	Feedback
RECEIVES	Charging Stations	1:N	Feedback
REQUIRES	Chargers	1:N	Maintenance
REPORTS	Users	1:N	Maintenance

Table 11: Cardinality Summary

5. Chen Notation ERD

The conceptual model that is expressed in standard notation is the Chen Notation ERD (Entity Relationship Diagram): the entities are represented by rectangles, relationships by diamonds, and attributes by ovals. Users entity centrally situated connected to several entities. There is hierarchical station infrastructure. The flow of booking involves Users, Vehicles and Reservations. The Reservations become Sessions to Payments. Maintenance lines and Maintenance lines.

Clear image available at: 🗎 ChargeAlaya Chen Notation ERD.png

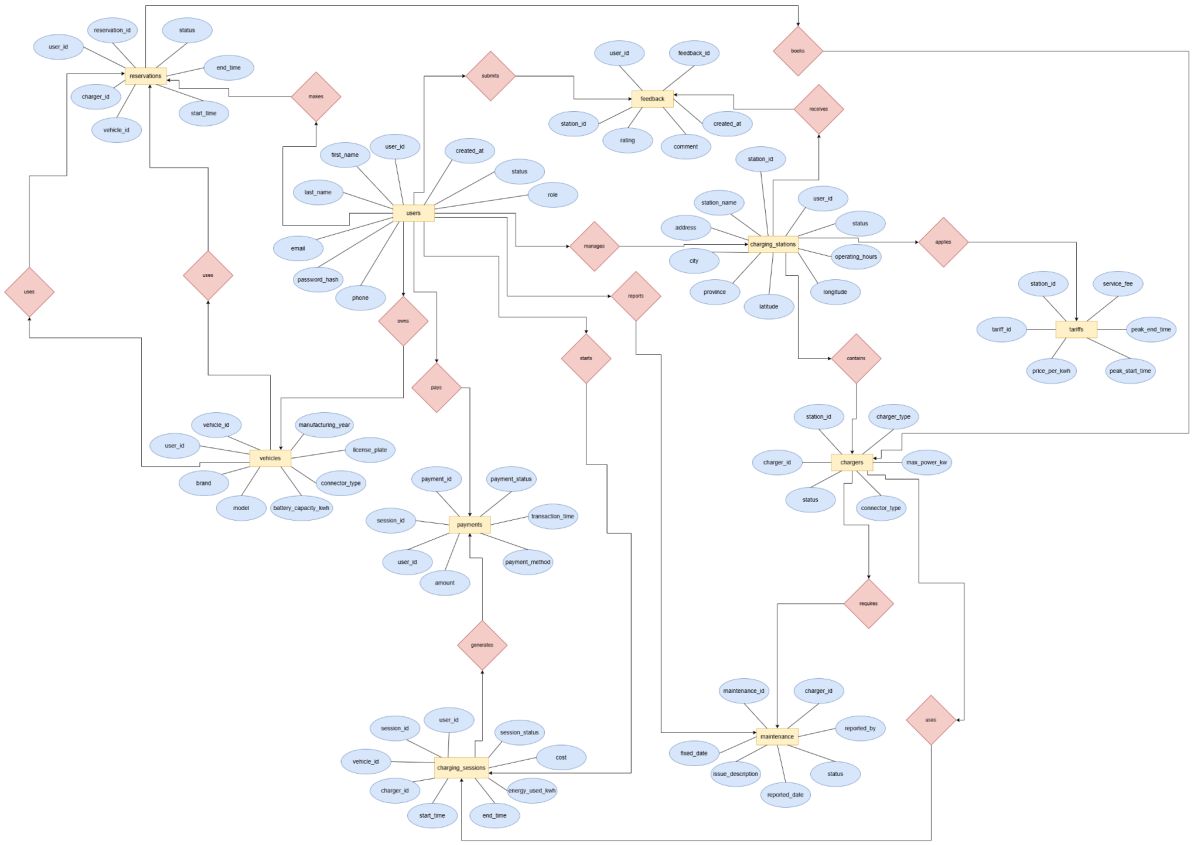

Fig. 2: Chen Notation ERD - ChargeAlaya

6. Physical ERD

Physical ERD is the one which demonstrates actual implementation with full specifications. Identified primary keys (PK) and foreign keys (FK).

Clear image available at: ChargeAlaya Physical ERD.png

Fig 3: Physical ERD - ChargeAlaya

7. Database Schema

7.1 Indexing Strategy

Primary keys that are auto-incremented are used as indexes which are clustered. Foreign keys are optimized using non-clustered indexes. Often indexed columns have been filtered.

Examples: There are indexes on users of id idx email and idx role on Chargers of id idx station status on Reservations of id idx reservation time and idx status on feedback of id idx station rating.

7.2 Referential Integrity

Integrity is enforced using foreign key constraints of CASCADE or SET NULL type. All records related to the user are also deleted. Chargers, tariffs, feedback Station deletion cascades. Session cascading Vehicle deletion to reservation SET NULL. Reservations, session, maintenance get deleted. The deletion of the sessions cascades to the payments.

8. Normalization

8.1 Normalization Level

ChargeAlaya schema is developed in a Third Normal Form (3NF), which guarantees data integrity with a minimum amount of redundancies and balances theory and performance.

8.2 First Normal Form Compliance

1NF presupposes atomic values, absence of any repetitive groups, and regular domains. All the requirements of the schema are adhered to: all columns contain single, indivisible values, there are no arrays or lists. operating hours are represented as a single string, and every attribute represents a unified domain.

Example: Each charger has one connector-type (e.g. "CCS2") instead of multiple connector-types (e.g. "CCS2, CHAdeMO, Type2”) (a violation of 1NF), which makes it easy to query.

8.3 Second Normal Form Compliance

NF 2NF presupposes 1NF and the absence of partial dependence. It is done through the utilization of single-column key primary keys, doing away with composite key and partial dependency. The non-key attributes are wholly dependent on the primary key.

Example: In the reservation, there are no attributes independent of the reservation, that is, all attributes (user_id, charger_id, vehicle_id, start-time, end-time, status) are based on the reservation-id.

8.4 Third Normal Form Compliance

3NF must have 2NF and no transitive dependencies. This is imposed in the schema by well defined separation of entities.

Station-Tariff separation: Price per kilowatt hour and service fee are separated off of charging station prices and into a separate tariffs table with a 1:1 relationship.

Maintenance-Charger separation: Maintenance information is kept separately allowing transitive dependencies to be eliminated, which allows complete recording of maintenance history.

8.5 Denormalization Decisions

Other values that are not stored are reservation cost, user full name, station average rating and the number of maintenance per charger among others since they are dynamically calculated or computed efficiently.

An exception that is made purposefully is charging_sessions.cost, where the final billing amount is recorded so that it is historically accurate and consistent and used in financial records.

9. Constraints and Business Rules

9.1 Database Constraints

9.1.1 Primary Key Constraints

All tables use AUTO_INCREMENT primary keys for unique identification.

9.1.2 Foreign Key Constraints

Sixteen foreign keys (fourteen 1:N, two 1:1) enforce referential integrity with CASCADE or SET NULL. Vehicles cascade on user_id. Reservations cascade on user_id and charger_id, while vehicle_id uses SET NULL. Tariffs and Payments maintain 1:1 relationships with Stations and Sessions respectively.

9.1.3 Unique Constraints

Users enforce unique email to prevent duplicate accounts and ensure reliable authentication.

9.1.4 Check Constraints

Feedback restricts ratings to 1-5 using CHECK (rating >= 1 AND rating <= 5).

9.1.5 NOT NULL Constraints

Users require names, email, phone, role; Chargers require charger_type and max_power_kw; Reservations require start_time and end_time.

9.1.6 DEFAULT Values

Users (role=ev_owner, status=active), timestamps (CURRENT_TIMESTAMP), Sessions (completed), Payments (pending), Chargers (available), Maintenance (open).

9.2 Business Rules Implementation

Eleven core business rules are enforced using constraints and application logic:

- end_time must be after start_time (application validation).
- Vehicle connector must match charger connector (reservation filtering).
- Chargers cannot have overlapping active reservations (conflict checks).
- Role-based access: station owners manage stations/sessions, admins manage accounts.
- Payments must be paid before session completion.
- Sessions default to completed.
- Payments limited to pending or paid.
- Users limited to admin, station_owner, ev_owner.
- Chargers limited to fast or normal.
- Maintenance status flows through open, in_progress, resolved.
- Feedback ratings restricted to 1-5.

9.3 Cascading Actions

Cascading rules prevent orphaned records. User deletion cascades to vehicles, reservations, sessions, payments, feedback, and maintenance. Station owner deletion cascades to stations. Station deletion cascades to chargers, tariffs, and feedback. Vehicle deletion cascades to sessions while reservations use SET NULL. Charger deletion cascades to reservations, sessions, and maintenance. Session deletion cascades to payments.

10. Conclusion

10.1 Summary

The ChargeAlaya ER model delivers a normalized, scalable database supporting EV charging operations with strong integrity, performance, security, and flexibility.

10.2 Key Achievements

The design includes ten core entities, sixteen relationships, full 3NF normalization, optimized indexing, enforced business rules, clear role separation, maintenance tracking, and a feedback system.

10.3 Design Strengths

Clear entity separation ensures scalability and consistency. Foreign keys prevent invalid references. Timestamps support auditing. Station-specific tariffs enable pricing flexibility. Stored session costs preserve financial accuracy. One-to-one session-payment mapping ensures correct billing. Maintenance and feedback support reliability and quality improvement.

10.4 Future Enhancements

Planned extensions include loyalty programs, real-time monitoring, analytics, multi-currency support, history exports, notifications, dynamic pricing, smart grid integration, and vehicle-to-grid support.

10.5 Implementation Readiness

The schema is production-ready with complete documentation, optimized performance, standards compliance, and a maintainable three-tier architecture.

10.6 Final Statement

ChargeAlaya's ER model provides a robust, scalable foundation for modern EV charging systems. It balances theoretical rigor with practical needs, ensures accurate modeling of one-to-one relationships, and supports secure, reliable growth.

Appendix: Glossary

ACID - Atomicity, Consistency, Isolation, Durability. A collection of attributes, which ensure that database transactions are done in a consistent manner.

CASCADE - A take place when a child record is described as referentially related to a parent record, when a change (an update or a delete) is made on the parent record, it is automatically propagated to the related child records in the system, ensuring referential integrity.

ERM - Entity Relationship Model. An abstract model that contains data that depicts entities, their features and how they are related to each other.

ERD - Entity Relationship Diagram. The Entity Relationship Model measured in standard notation.

FK - Foreign Key.A column or a group of columns of a table, which is pointed to the primary key of another table, and create relationships between tables.

InnoDB - This is a MySQL storage engine which supports ACID-compliant transactions, foreign key constraints and crash recovery.

Normalization - It means the arrangement of database tables and columns to minimize data duplication and enhance the data integrity.

1NF - First normal form A normalization of a database where all attributes include atomic (indivisible) values, and no repeat groups are present.

2NF - Second Normal Form. A degree of normalization that necessitates 1NF compliance and also removal of partial dependencies on the primary key.

3NF - Third Normal Form. A normalization level that mandates the 2NF compliance and removal of transitive dependencies among the non-key attributes.

PK - Primary Key. A column or columns consisting of a unique identifier of the rows in a table.

RBAC - Role-Based Access Control. A security mechanism, where access to systems is limited to the user roles (admin, station-owner, ev-owner).

Referential Integrity - This is a database constraint that is used to provide integrity between rows within the database tables so that no record is orphaned.

SET NULL - A referential option that alters foreign key to the value of NULL in case the record to which a foreign key refers is deleted and historical information is maintained.

UTF8MB4 - UTF8MB4 is a Character encoding that supports the complete Unicode character set, Emojis and special characters.